Dosage & IV Drip Rates

Charter demo: every component on one sheet

MindShelf

1 Core formulas

Desired over have

$$V = \frac{D}{H} \times Q$$

D ordered dose, H dose on hand, Q volume containing H.

Gravity drip rate

$$\text{gtt/min} = \frac{V_{\text{mL}} \times \text{DF}_{\text{gtt/mL}}}{t_{\text{min}}}$$

$$\text{mL/h} = \frac{V_{\text{mL}}}{t_{\text{h}}} \qquad 1\ \text{mg} = 1000\ \mu\text{g}$$

EXAMPLE Oral suspension

Ordered 250 mg; on hand 500 mg per 5 mL.

$$V = \frac{250}{500} \times 5 = 2.5\ \text{mL}$$

EXAMPLE Gravity infusion

1000 mL over 8 h, drop factor 15 gtt/mL.

$$\frac{1000 \times 15}{480} = 31.25 \Rightarrow 31\ \text{gtt/min}$$

TRAP

Convert **before** you divide: mixing mg and µg multiplies the error by 1000.

TRAP Rounding

Round only the final result, never the intermediate steps.

2 Reference values

Set	Drop factor	Typical use
Macro-drip	10–20 gtt/mL	Adults, large volumes
Micro-drip	60	Pediatrics, small volumes
Blood set	10–15 gtt/mL	Blood products

2.1 Number formatting

- Helper `nombre()`: 1,234,567.89, −0.5, 12,500, 4.0
- Tabular figures keep columns aligned.
  - Second-level item.

CLINICAL NOTE

Two independent checks for high-alert drugs (insulin, heparin, potassium).

Method in 3 steps

1. Write the formula with units.
2. Convert to the same unit.
3. Check the order of magnitude.

2.1.1 Level-3 heading

Body text in Fira Sans, set ragged-right for narrow columns. A external link is underlined; **bold** is semi-bold.

3 Three-column block

Unit	Equals
1 g	1000 mg
1 mg	1000 µg
1 L	1000 mL
1 h	60 min

EXAMPLE Weight-based dose

Ordered 15 mg/kg; weight 12 kg.

$$15 \times 12 = 180\ \text{mg}$$

TRAP Zeros

No trailing zero (5.0 mg is misread as 50 mg); always a leading zero (0.5 mg, not .5 mg).

4 Full-width table

Volume	Duration	mL/h	gtt/min (20)	Comment
500 mL	4 h	125	42	Rounded from 41.7
1000 mL	8 h	125	42	Same hourly rate
250 mL	30 min	500	167	Check the pump limit
100 mL	20 min	300	100	Short infusion

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